

Generative AI (22ADC14)

Assignment - 1

B.E. VII Semester, Section – I1

Submission Date: 05-09-2026

Case Studies on Autoencoders, Variational Autoencoders, AutoRegressive Models and Variants

General Instructions

1. **Team Allocation** Each team has been pre-assigned one case study from the given list.
2. **Submission Format** Submissions must strictly follow the Case Study Submission Template provided in this assignment document.
3. **Submission Mode** Submissions must be made in hard copy format only.

Case Studies

Case Study 1: Autoencoder for Image Denoising [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Add Gaussian noise to MNIST or FashionMNIST and train a basic autoen-coder.
- **Task 2** Observe reconstruction quality at different noise levels.
- **Task 3** Evaluate if such an autoencoder could generalize to natural noisy images.

Team Details

- **Team 1:** Roll Nos. 1601-23-771-001, 1601-23-771-002, 1601-23-771-003
- **Team 2:** Roll Nos. 1601-23-771-004, 1601-23-771-005, 1601-23-771-006

Case Study 2: Sparse Autoencoder on Student Marks Dataset [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Train a sparse AE on student subject marks (Math, Coding, Physics).
- **Task 2** Show how sparsity highlights hidden skills or interests.
- **Task 3** Evaluate if sparse AE is better than plain AE for interpretability in educational data.

Team Details

- **Team 1:** Roll Nos. 1601-23-771-007, 1601-23-771-008, 1601-23-771-009
- **Team 2:** Roll Nos. 1601-23-771-010, 1601-23-771-011, 1601-23-771-012

Case Study 3: Variational Autoencoder for Anomaly Detection [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Train a VAE on FashionMNIST normal classes and generate reconstructions.
- **Task 2** Compare reconstruction quality of normal vs anomalous samples.
- **Task 3** Evaluate performance against a basic autoencoder.

Team Details

- **Team 1:** Roll Nos. 1601-23-771-013, 1601-23-771-014, 1601-23-771-015
- **Team 2:** Roll Nos. 1601-23-771-016, 1601-23-771-017, 1601-23-771-018

Case Study 4: FVSBN for Weather Time-Series Forecasting [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Train Fully Visible Sigmoid Belief Network (FVSBN) on a small weather dataset.
- **Task 2** Analyze how FVSBN models sequential dependencies.
- **Task 3** Evaluate predictions against a simple RNN baseline.

Team Details

- **Team 1:** Roll Nos. 1601-23-771-019, 1601-23-771-020, 1601-23-771-021
- **Team 2:** Roll Nos. 1601-23-771-022, 1601-23-771-023, 160123771024

Case Study 5: NADE for Text Modeling [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Train NADE on a binary bag-of-words dataset.
- **Task 2** Analyze how NADE captures word dependencies.
- **Task 3** Evaluate scalability for large vocabulary datasets.

Team Details

- **Team 1:** Roll Nos. 160123771025, 160123771026, 160123771027
- **Team 2:** Roll Nos. 160123771028, 160123771029, 160123771030

Case Study 6: MADE on Stock Price Sequences [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Train MADE on simplified stock market trend dataset.
- **Task 2** Analyze how masking ensures correct dependency order.
- **Task 3** Compare performance with a GRU/LSTM model.

Team Details

- **Team 1:** Roll Nos. 160123771031, 160123771032, 160123771033
- **Team 2:** Roll Nos. 160123771034, 160123771035, 160123771036

Case Study 7: Comparison of VAE, Sparse AE, and MADE [BL-4 5] [CO1,CO2,CO3]

- **Task 1** Train simple models (VAE, Sparse AE, MADE) on MNIST.
- **Task 2** Compare latent features (VAE: diversity, Sparse AE: compressed, MADE: dependencies).
- **Task 3** Evaluate which model is most practical for a student project.

Team Details

- **Team 1:** Roll Nos. 160123771037, 160123771039, 160123771040
- **Team 2:** Roll Nos. 160123771041, 160123771042, 160123771044
- **Team 3:** 160123771305, 160123771306, 160123771307

Case Study 8: Most existing GAN-based approaches for plant disease image generation treat the process as a black-box mapping and lack mechanisms to control or interpret which visual and environmental factors influence the generated symptoms. This limits trust and practical utility in precision agriculture. An attention-guided multimodal GAN can explicitly model the interaction between environmental conditions and localized disease symptoms, producing high-fidelity, controllable, and more interpretable synthetic samples. **[BL-4 5]**
[CO1,CO2,CO3]

Dataset Link:

https://drive.google.com/file/d/1DQjOB3gPdOU9jnqpcDO4sfgpfTEG-5x6/view?usp=drive_link

Task1: Design and develop an attention-guided multimodal GAN that incorporates environmental sensor data as conditioning information and simultaneously generates high-quality disease images along with spatially meaningful attention maps.

Task 2: Evaluate the generation quality, fidelity, and controllability of disease symptoms under varying environmental conditions using both quantitative metrics and qualitative analysis.

Task 3 : Assess the interpretability of the learned attention mechanisms through visualization of attention maps and quantitative localization metrics.

Task 4: Investigate the practical usefulness of the generated synthetic samples and attention maps in improving downstream disease classification accuracy and model explainability.

Team Details

- **Team 1:** Roll Nos. 160123771045, 160123771047, 160123771048
- **Team 2:** Roll Nos. 160123771049, 160123771050, 160123771051
- **Team 3:** Roll Nos. 160123771052, 160123771053, 160123771054

- **Team 4:** 160123771064, 160123771065, 160123771301

Case Study 9: Multimodal Generative Representation Learning for Plant Disease Analysis [BL-4 5] [CO1,CO2,CO3]

Existing plant disease detection systems predominantly rely on **leaf images alone**, which may not adequately represent disease-related information such as crop type, disease symptoms, severity, and textual descriptions. Furthermore, limited and imbalanced samples for certain diseases can affect the ability of deep learning models to learn robust representations. Therefore, this research aims to develop a **multimodal generative representation learning framework** that integrates plant leaf images with associated textual and metadata information using **Autoencoders and Variational Autoencoders (VAEs)** to learn compact and meaningful representations of plant diseases.

Dataset Link:

https://drive.google.com/file/d/1DQjOB3gPdOU9jnqpcDO4sfpgfTEG-5x6/view?usp=drive_link

Tasks

Tasks 1: Multimodal Dataset Construction:

Develop or prepare a multimodal dataset containing plant leaf images, disease classes, crop information, symptom descriptions, and other relevant metadata.

Task 2 : Multimodal Representation Learning:

Develop an Autoencoder/VAE-based architecture to learn latent representations from the multimodal plant disease information.

Task 3 : Representation Comparison:

Compare basic Autoencoder, Regularized Autoencoder, and VAE representations in terms of reconstruction quality and latent-space separability.

Task 4 : Performance Evaluation:

Evaluate the proposed approach using reconstruction loss, classification performance, latent-space visualization, and other suitable metrics.

Team Details

- **Team 1:** Roll Nos. 160123771055, 160123771056, 160123771057
- **Team 2:** Roll Nos. 160123771058, 160123771059, 160123771060
- **Team 3:** Roll Nos. 160123771061, 160123771062, 160123771063
- **Team 4:** Roll Nos. 160123771302, 160123771303, 160123771304

Case Study Submission Template

Team Details

- Team Name:
- Team Members (Roll Nos. & Names):
 1.
 2.
 3.
 4.

Case Study Title

(e.g., Autoencoder for Image Denoising)

1. Problem Statement (1 mark)

Briefly describe the real-world problem (2–3 sentences).

2. Implementation Setup (2 marks)

- Dataset used
- Model type
- Limited implementation details (epochs, layers, activation)

3. Observations / Results (2 marks)

- Output screenshots or plots
- Short description of key observations

4. Analysis (Bloom's L4) (2.5 marks)

- Patterns and comparisons
- Explanation of what the model learned

5. Evaluation (Bloom's L5) (2.5 marks)

- Strengths and weaknesses
- Feasibility for larger datasets
- Suggestions for improvement

6. Conclusion (0.5 mark)

Summarize what was learned in 2–3 lines.